

Marking Scheme for Project #1				
Criterion	Excellent	Average	Below Average	Score (20 Points)
<u>Circuit One (10 points)</u>				
Circuit - Diagram and KVL/KCL Derivation	circuit diagram with proper annotations is used to derive KVL/KCL equations (2 points)	circuit diagram lacks annotations, or some of the KVL/KCL equations are not adequately presented (1 point)	circuit diagram and most of the KCL/KVL equations are not presented (0 point)	
KVL/KCL Equations - Matrix form with proper pivoting	equations presented in the final matrix form with partial pivoting (2 points)	equations presented in the final matrix form with no partial pivoting (1 point)	equations were not presented in matrix form (0 point)	
Gauss Seidel - Programming	code utilized error criterion, and demonstrated MATLAB fundamental knowledge (loops, arrays and conditions). (2 points)	code utilized error criterion, but did not demonstrate MATLAB fundamental knowledge (loops, arrays and conditions) (1 point)	code did not utilize error criterion, and did not demonstrate MATLAB fundamental knowledge (loops, arrays and conditions) to find the roots. (0 point)	
Gauss Seidel – Relaxation and Solution	code applied relaxation and could find the correct roots. (2 points)	code applied relaxation, but the correct roots were not found. (1 point)	code did not apply relaxation, and terminated with an error (0 point)	
Discussion – convergence, behaviour with under and over relaxation, and influence of relaxation on number of iterations	un-relaxed, under-relaxed and over-relaxed solutions are discussed reporting the effect of relaxation on the solution convergence and number of iterations (2 points)	un-relaxed, under-relaxed or over-relaxed solutions was not discussed, or the effect of relaxation on the solution convergence or number of iterations was not reported (1 point)	poor or no discussion on the solutions and the influence of relaxation parameter on number of iterations and convergence. (0 point)	
<u>Circuit Two (10 points)</u>				
Circuit - Diagram and KVL/KCL Derivation	circuit diagram with proper annotations is used to derive KVL/KCL equations (2 points)	circuit diagram lacks annotations, or some of the KVL/KCL equations are not adequately presented (1 point)	circuit diagram and most of the KCL/KVL equations are not presented (0 point)	
KVL/KCL Equations - Matrix form with proper pivoting	equations presented in the final matrix form with partial pivoting (2 points)	equations presented in the final matrix form with no partial pivoting (1 point)	equations were not presented in matrix form (0 point)	
Gauss Seidel - Programming	code utilized error criterion, and demonstrated MATLAB fundamental knowledge (loops, arrays and conditions). (2 points)	code utilized error criterion, but did not demonstrate MATLAB fundamental knowledge (loops, arrays and conditions) (1 point)	code did not utilize error criterion, and did not demonstrate MATLAB fundamental knowledge (loops, arrays and conditions) to find the roots. (0 point)	
Gauss Seidel – Relaxation and Solution	code applied relaxation and could find the correct roots. (2 points)	code applied relaxation, but the correct roots were not found. (1 point)	code did not apply relaxation, and terminated with an error (0 point)	
Discussion – convergence, behaviour with under and over relaxation, and influence of relaxation on number of iterations	un-relaxed, under-relaxed and over-relaxed solutions are discussed reporting the effect of relaxation on the solution convergence and number of iterations (2 points)	un-relaxed, under-relaxed or over-relaxed solutions was not discussed, or the effect of relaxation on the solution convergence or number of iterations was not reported (1 point)	poor or no discussion on the solutions and the influence of relaxation parameter on number of iterations and convergence. (0 point)	
Total				/20